Mini-project 6.3

Customer segmentation with clustering

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Contents

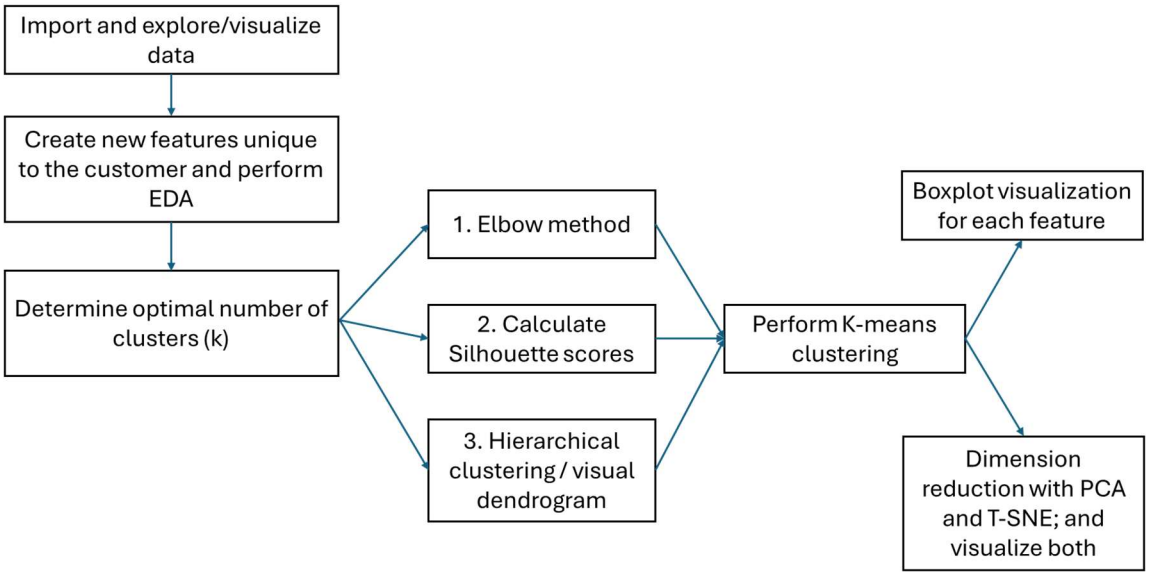
1.0 Overview 3
2.0 Approach 3
2.1 EDA 3
2.1.1 Feature creation 4
2.1.2 Visualise new features 5
2.2 Clustering with machine learning methods 6
2.2.1 Elbow 6
2.2.2 Silhouette 7
2.2.3 Hierarchical 7
2.2.4 Optimal K 8
2.2.5 K-means clustering 8
2.3 Visualisation with PCA and T-SNE 9
2.3.1 PCA 9
2.3.2 T-SNE 9
3.0 Conclusion 10
References 10

1.0 Overview

The project requires customer segmentation from the e-commerce data provided (SAS, 2024). Being able to segment the customers data by clustering models, will help the business to provide better customer centric improvements to marketing.

2.0 Approach

The data provided was imported and exploratory data analysis (EDA) was carried out, including feature visualisation. Three different techniques were used for to determine the optimal number of clusters (k). Once k was known, k-means clustering could be performed. The clusters were then visualised in 3 different methods.



The data imported showed 20 features with 951,669 rows of data (snippet below).

```
# View the DataFrame and perform EDA, identify missing values, duplication
print(data.shape)

# Display the first few rows
data.head()
```

```
(951669, 20)
```

	Quantity	City	Continent	Postal_Code	State_Province	Order_Date	Delivery_Date	Total Revenue	Unit Cost	Discount	OrderTypeLabel	CustomerCountryLabel	Customer_BirthDate
0	3	Leinster	Oceania	6437	Western Australia	01JAN2012	07JAN2012	$28.50	$9.10	.	Internet Sale	Australia	08MAY1978
1	2	Berowra	Oceania	2081	New South Wales	01JAN2012	04JAN2012	$113.40	$56.90	.	Internet Sale	Australia	13DEC1978
2	2	Berowra	Oceania	2081	New South Wales	01JAN2012	04JAN2012	$41.00	$18.50	.	Internet Sale	Australia	13DEC1978
3	1	Northbridge	Oceania	2063	New South Wales	01JAN2012	03JAN2012	$35.20	$29.60	.	Internet Sale	Australia	22JUN1997
4	1	Montréal	North America	NaN	Quebec	01JAN2012	04JAN2012	$24.70	$23.60	.	Internet Sale	Canada	28JAN1978

2.1 EDA

Missing values on 3 features and 21 duplicate rows were identified.

Different data types in the features including 5 integers/numeric columns. Note - columns such as 'Total Revenue' is not an int64/numeric feature.

Missing values on 3 features: city (135 rows), postal_code (3716 rows) and state_province (117,192 rows).

```
# Determine metadata of DataFrame.
data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 951669 entries, 0 to 951668
Data columns (total 20 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   Quantity              951669 non-null  int64
 1   City                  951534 non-null  object
 2   Continent             951669 non-null  object
 3   Postal_Code           947953 non-null  object
 4   State_Province        834477 non-null  object
 5   Order_Date            951669 non-null  object
 6   Delivery_Date         951669 non-null  object
 7   Total Revenue         951669 non-null  object
 8   Unit Cost             951669 non-null  object
 9   Discount              951669 non-null  object
 10  OrderTypeLabel        951669 non-null  object
 11  CustomerCountryLabel  951669 non-null  object
 12  Customer_BirthDate    951669 non-null  object
 13  Customer_Group        951669 non-null  object
 14  Customer_Type         951669 non-null  object
 15  Order ID              951669 non-null  int64
 16  Profit                951669 non-null  object
 17  Days to Delivery      951669 non-null  int64
 18  Loyalty Num           951669 non-null  int64
 19  Customer ID           951669 non-null  int64
dtypes: int64(5), object(15)
memory usage: 145.2+ MB
```

```
# Find missing values.
data.isnull().sum()
```

	0
Quantity	0
City	135
Continent	0
Postal_Code	3716
State_Province	117192
Order_Date	0
Delivery_Date	0
Total Revenue	0
Unit Cost	0
Discount	0
OrderTypeLabel	0
CustomerCountryLabel	0
Customer_BirthDate	0

```
# Check for duplicate values
duplicate = data[data.duplicated()]

print("Duplicate Rows : ", duplicate.count())

# Print the resultant Dataframe
duplicate

Duplicate Rows :  Quantity              21
City                  21
Continent             21
Postal_Code           21
State_Province        21
Order_Date            21
Delivery_Date         21
Total Revenue         21
Unit Cost             21
Discount              21
OrderTypeLabel        21
CustomerCountryLabel  21
Customer_BirthDate    21
Customer_Group        21
Customer_Type         21
Order ID              21
Profit                21
Days to Delivery      21
Loyalty Num           21
Customer ID           21
dtype: int64
```

21 duplicate rows

	Quantity	City	Continent	Postal_Code	State_Province	Order_Date	Delivery_Date	Total Revenue	Unit Cost	Discount	OrderType
25809	1	Köln	Europe	51105	Nordrhein-Westfalen	08MAR2012	08MAR2012	$59.70	$59.90	.	Retu
59723	2	Roma	Europe	00167	Lazio	24MAY2012	24MAY2012	$26.40	$11.90	.	Retu
60475	1	Madrid	Europe	28042	Madrid, Comunidad de	26MAY2012	29MAY2012	$55.10	$55.30	.	Catalo

The null values weren't provided by customers during the order and might not be mandatory. The 21 duplicate transactions might be valid. Hence this data wasn't removed (potential question to domain expert).

Descriptive satistics shows:

```
data.describe()
```

	Quantity	Order ID	Days to Delivery	Loyalty Num	Customer ID
count	951669.000000	9.516690e+05	951669.000000	951669.000000	951669.000000
mean	1.678438	1.236943e+09	1.056107	66.534079	45440.602683
std	0.899914	4.273202e+06	2.065725	46.476795	27214.145206
min	1.000000	1.230000e+09	0.000000	0.000000	1.000000
25%	1.000000	1.233081e+09	0.000000	0.000000	22139.000000
50%	1.000000	1.236755e+09	0.000000	99.000000	44140.000000
75%	2.000000	1.240689e+09	1.000000	99.000000	70411.000000
max	10.000000	1.244338e+09	32.000000	99.000000	94254.000000

Numeric fields shown here only. Some are irrevelant to use eg order ID.

2.1.1 Feature creation

New features were created to help marketing understand the customer base. Order transaction data were converted to customer marketing metric data (Data Science, 2025).

New features	How was it created?	How was it aggregated?
Frequency	Count of 'Customer ID'	Maximum
Recency	Today's date minus the Delivery Date in days	Minium
Customer Lifetime Value (CLV)	Existing field 'Total Revenue' in numeric format	Sum
Average Unit Cost	Existing field 'Unit Cost' in numeric format	Mean
Customer age	Today's date minus the date of birth in years	Maximum

New features were created, aggregated (above) and grouped by customers, making it a customer-centric data required for marketing.

	Customer ID	Frequency	Recency	CLV	Age	Avg_Unit_Cost
0	1	35	3320	4322.70	72	89.622857
1	3	20	3422	2821.00	42	90.505000
2	4	9	4411	782.67	42	37.195556
3	5	19	3308	2810.98	37	107.947368
4	6	15	3493	1604.58	67	52.526667

EDA on new features:

The new aggregated dataset has 68,299 rows. Less than original dataset due to aggregation. We have one row per customer. All numeric features

```
df_aggr.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 68300 entries, 0 to 68299
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Customer ID    68300 non-null  int64
 1   Frequency      68300 non-null  int64
 2   Recency        68300 non-null  int64
 3   CLV            68300 non-null  float64
 4   Age            68300 non-null  int64
 5   Avg_Unit_Cost  68300 non-null  float64
dtypes: float64(2), int64(4)
memory usage: 3.1 MB
```

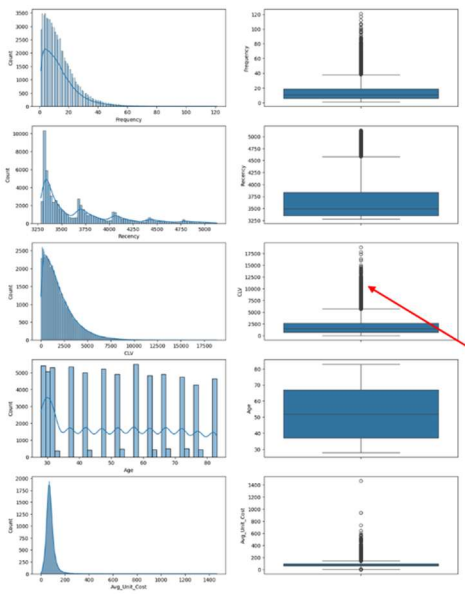
The new features added are numeric. Frequency goes from 1 to 121, but its mean is 13, showing its skewed towards the left. This is similar for Recency, CLV and Avg_Unit_Cost

```
df_aggr.describe()
```

	Customer ID	Frequency	Recency	CLV	Age	Avg_Unit_Cost
count	68300.000000	68300.000000	68300.000000	68300.000000	68300.000000	68300.000000
mean	45447.623397	13.933660	3669.399971	1950.168370	52.184802	78.894765
std	27203.855105	11.329121	414.289931	1719.939245	17.482617	38.004880
min	1.000000	1.000000	3283.000000	0.630000	28.000000	0.500000
25%	22160.500000	6.000000	3347.000000	696.000000	37.000000	57.579792
50%	44140.500000	11.000000	3491.000000	1497.450000	52.000000	73.449286
75%	70423.250000	19.000000	3841.000000	2709.845000	67.000000	92.515260
max	94254.000000	121.000000	5127.000000	18860.960000	83.000000	1463.500000

2.1.2 Visualise new features

Plotting new features in a box-plot/histogram shows distribution.



Frequency, CLV and Avg unit cost (are skewed to the left) as expected and has a number of outliers seen on the box plot chart.

Recency is also left skewed but less curve like and have more peaks in-between, though smaller after the initial most left peak. There are also outliers here.

Age have a number of variation across the different ages - either really high or low. No outliers seen.

Outliers

Example: Calculating inter-quartile range on ‘Recency’ to view anomalies shows 3,353 rows, out of 68,299 (~ 5%)

```
IQR calculations for Recency is below:
The value of Q1 is: 3347.0
The value of Q3 is: 3841.0
The value of IQR is: 494.0
Upper limit:  4582.0
Lower limit:  2606.0
Shape is :  (3353, 7)
```

	Customer ID	Frequency	Recency	CLV	Age	Avg_Unit_Cost	Recency IQR
52	68	2	4934	569.2	82	89.600000	1
84	107	1	5095	133.2	62	60.600000	1
114	150	11	4765	1180.1	42	68.090909	1
119	158	3	4771	135.5	82	37.766667	1
213	283	1	4936	76.2	62	33.100000	1

2.2 Clustering with machine learning methods

Scaled data (standard scaler method used) is required for k-means clustering (so that priority isn’t given to larger features).

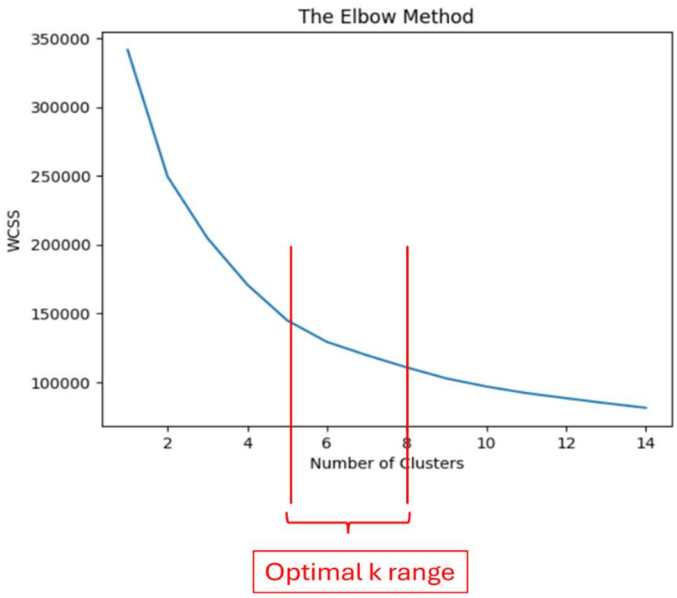
```
# View output.
data_scaled_only.head()
```

	Frequency SC	Recency SC	CLV SC	Age SC	Avg_Unit_Cost SC
0	1.859499	-0.843377	1.379438	1.133431	0.282284
1	0.535468	-0.597171	0.506319	-0.582572	0.305496
2	-0.435488	1.790064	-0.678807	-0.582572	-1.097215
3	0.447199	-0.872342	0.500493	-0.868572	0.764450
4	0.094124	-0.425792	-0.200932	0.847431	-0.693813

To perform k-means clustering, we need to find the optimal k.

2.2.1 Elbow

Graph plotted for range of 1-15 clusters. The reduction of the graph reduces at ~5-8 suggesting a range of optimal k values.

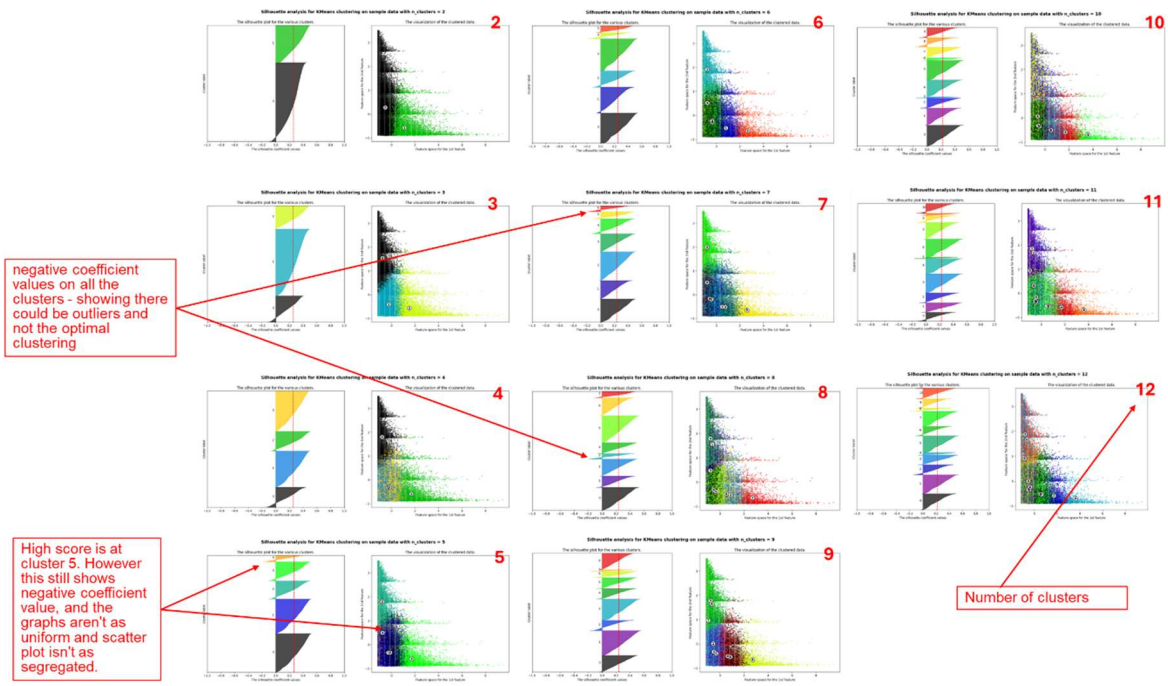


2.2.2 Silhouette

This method provides silhouette scores, with optimal scores being closest to 1 for optimal k. Scores were calculated for 2-12 clusters. Clusters of 5 has the highest score (0.267). Top scores are for optimal k being between 2-5.

Number of Clusters	Silhouette Score
2	0.2603017587026929
3	0.25257602065484536
4	0.25315414746484477
5	0.26703173156456056
6	0.25253544739480965
7	0.23512371379172697
8	0.23576973833204393
9	0.2403065463502543
10	0.22535935401728513
11	0.2271958310179285
12	0.22253420062923862

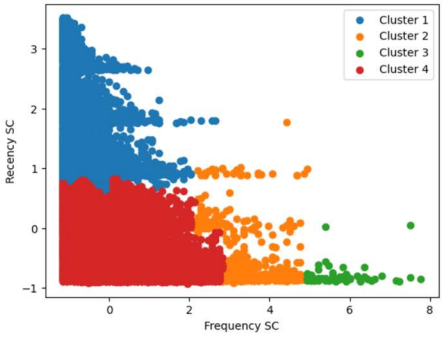
Visualising clusters (notebook for zoomed in graph).

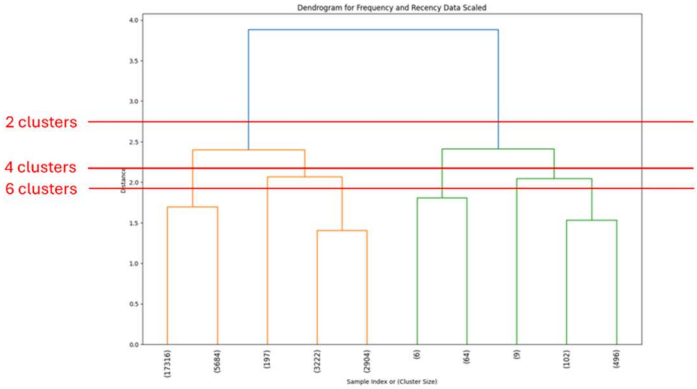


Ideal scenario would be no scores of < 0 and stable distribution of coefficient. Distributions are similar for clusters 3,4,5.

2.2.3 Hierarchical

Performing Hierarchical Agglomerative Clustering for 5 clusters (common to silhouette and elbow method) produces the below scatter plot (Used 2 features: Frequency and Recency) and dendrogram.





Optimal k (clusters) from the dendrogram potentially could be: 2, 4 or 6.

2.2.4 Optimal K

Finding the overlapping optimal k with the 3 methods isn't possible:

Elbow	Silhouette	Hierarchical
5	3	2
6	4	4
7	5	6
8		

Finding overlapping k with 2 methods is possible: optimal-k 4, 5, 6. Selecting silhouette provides a numerical view. Both hierarchical and elbow are subjective as are visual. Choosing clusters 4 or 5 would be suitable as hierarchical is easier to read, compared to elbow. Optimal k is selected as 4.

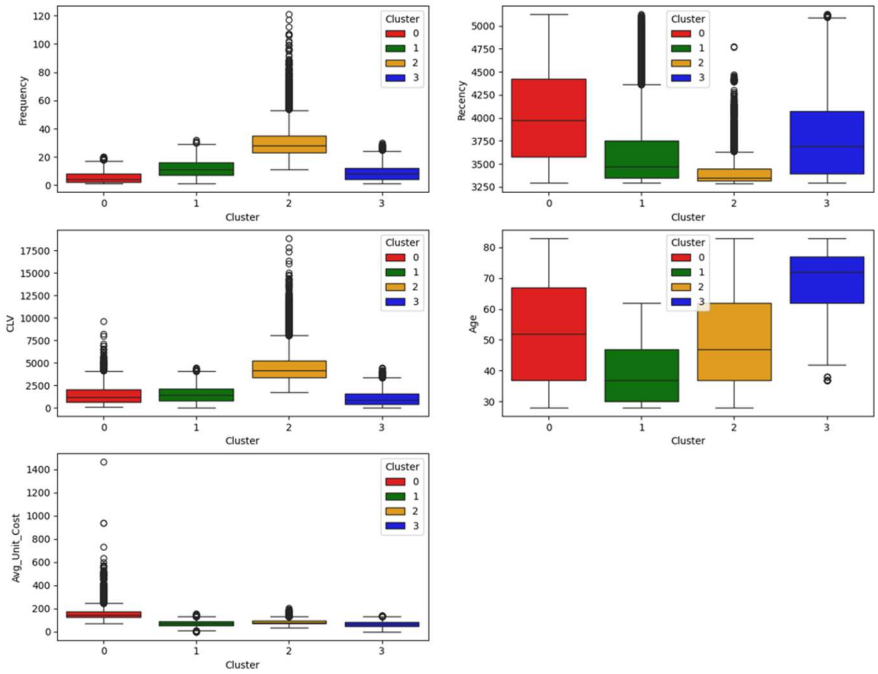
2.2.5 K-means clustering

Performed k-means clustering on scaled data for k=4. The cluster allocations (last column) is merged with the original data to provide a cluster per customer.

```
# Set the Number of clusters
n_clusters = 4

# Perform KMeans clustering. KMeans required scaled data
kmeans = KMeans(n_clusters=n_clusters, random_state=0).fit(data_scaled_only)

# Assign clusters back to the original DataFrame
df_aggr_orig['Cluster'] = kmeans.labels_

df_aggr_orig.head()
```

	Customer ID	Frequency	Recency	CLV	Age	Avg_Unit_Cost	Cluster
0	1	35	3320	4322.70	72	89.622857	2
1	3	20	3422	2821.00	42	90.505000	1
2	4	9	4411	782.67	42	37.195556	1
3	5	19	3308	2810.98	37	107.947368	1
4	6	15	3493	1604.58	67	52.526667	3

Visualising for each feature:



Interpretability is possible eg: Cluster 0 indicates lower frequency values, middle range of Recency, lower CLV, middle range for age and higher avg unit cost. CLV and avg unit cost both have outliers.

2.3 Visualisation with PCA and T-SNE

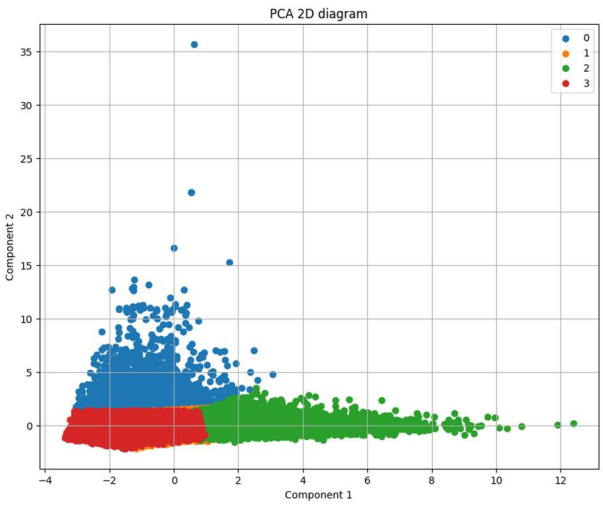
2.3.1 PCA

PCA reduces 5 features to 2, providing a 2D scatter plot based on the k-means clustering data with 4 clusters.

```
# applying PCA

pca = PCA(n_components=2) #reducing to two features
pca_model = pca.fit(data_scaled_only)

df_trans = pd.DataFrame(pca_model.transform(data_scaled_only),
                        columns=['pca1', 'pca2'])
df_trans
```

	pca1	pca2
0	2.183836	0.063622
1	1.038260	0.170730
2	-1.430575	-0.700607
3	1.187745	0.559341
4	-0.104173	-0.798344
...	...	...
68295	-1.042015	0.570763
68296	-1.141359	1.225505
68297	4.925102	0.031401
68298	-0.556730	-0.116798
68299	-1.346465	0.603228

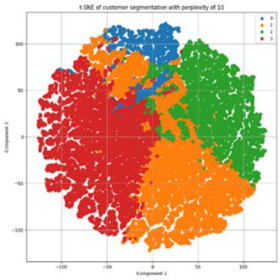
68300 rows × 2 columns



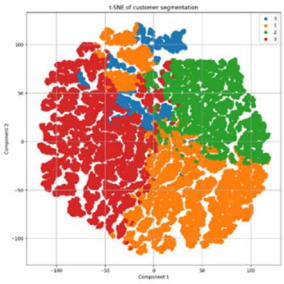
The orange colour (cluster 1) have very few points which might overlap with the red clusters for example. There are a number of outliers seen as cluster 0 (blue) at the top of the graph, and also as cluster 2 (green) on the bottom right outliers.

2.3.2 T-SNE

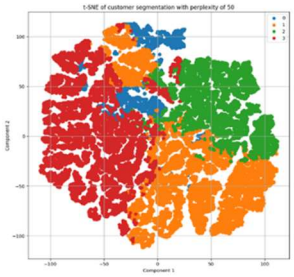
TSNE visualisation was used with different perplexity numbers of 10,30,50.



Perplexity as 10



Perplexity as 30



Perplexity as 50

Perplexity 10 shows a more compact view. The clusters (different colours) are intertwined with boundaries of each other - with the orange data points (cluster 1) going through the middle and overlapping with the other clusters.

As it shifts to perplexity 30 and 50, the clusters are spread out and less compact. Blue is least visible (category 0). This might be because of overlaps. Perplexity 10 seem to be the most formed compact clusters in comparison to the others.

3.0 Conclusion

Three different techniques (elbow, silhouette and hierarchical) were used to identify the optimal k, by comparing the outputs and looking at commonalities. Visualisation helps to see how the data points are spread out spatially and understand the data better. This provides a view for marketing on customer segmentation.

PCA and T-SNE was used for 2D plots. T-SNE took much longer (>20 mins) due to its non-linear algorithm computational cost, providing options to zoom by varying perplexity. The data can be sampled to reduce the time taken. The 4 clusters are more clearer with T-SNE.

The downside to both PCA and T-SNE is interpretability. Boxplots were used for each feature to show what each of the clustering translate back to in relation to data and features, helping the marketing team.

Next steps would be to discuss any additional data that can be provided by the domain expert, which can help segment the data further or provide additional context.

References

SAS, 2024. CUSTOMERS_CLEAN [Data set]. SAS. Last revised on 15 December 2021. [Accessed: 20 February 2024].

(from Cambridge data science accelerator course material)

Data Science, 2025. *Data Science with Machine Learning & AI Career Accelerator* programme at the University of Cambridge [Accessed: 7 January 2026]